

CDAC MUMBAI
LAB ASSIGNMENT 4

Question 1: Print Numbers from 1 to N

Problem Statement:

Write a Java program that asks the user for a number N and then prints the numbers from 1 to N using a for loop.

Sample Input: Enter a number: 10

Expected Output: 1 2 3 4 5 6 7 8 9 10

Java Code:

```
import java.util.Scanner;
public class PrintNumbers {
    public static void main(String[] args) {
        Scanner sc= new Scanner(System.in);

        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        for (int i = 1; i <= n; i++) {
            System.out.print(i + " ");
        }

        sc.close();
    }
}
```

Output:

PS F:\Java Code\Logic Building\Assignment 4> javac PrintNumbers.java

PS F:\Java Code\Logic Building\Assignment 4> java PrintNumbers

Enter a number: 10

1 2 3 4 5 6 7 8 9 10

PS F:\Java Code\Logic Building\Assignment 4>

Question 2: Print Multiples of 3 between 1 and N

Problem Statement:

Write a Java program that asks the user for a number N and prints all the multiples of 3 between 1 and N using a for loop.

Sample Input: Enter a number: 20

Expected Output: 3 6 9 12 15 18

Java Code:

```
import java.util.Scanner;

public class MulOfThree {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        for (int i = 3; i <= n; i = i + 3) {
            System.out.print(i + " ");
        }
        sc.close();
    }
}
```

Output:

PS F:\Java Code\Logic Building\Assignment 4> java MulOfThree

Enter a number:30

3 6 9 12 15 18 21 24 27 30

PS F:\Java Code\Logic Building\Assignment 4>

Question 3: Calculate the Factorial of a Number

Problem Statement:

Write a Java program that asks the user for a number N and calculates the factorial of N using a for loop.

Sample Input: Enter a number: 5

Expected Output: Factorial of 5 is 120

Java Code:

```
import java.util.Scanner;

public class Factorial {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        long factorial = 1;

        for (int i = 1; i <= n; i++) {
            factorial = factorial * i;
        }

        System.out.println("Factorial of " + n + " is " + factorial);
        sc.close();
    }
}
```

Output:

PS F:\Java Code\Logic Building\Assignment 4> javac Factorial.java

PS F:\Java Code\Logic Building\Assignment 4> java Factorial

Enter a number: 5

Factorial of 5 is 120

PS F:\Java Code\Logic Building\Assignment 4>

Question 4: Print Even Numbers from 1 to N

Problem Statement:

Write a Java program that asks the user for a number N and prints all the even numbers from 1 to N using a for loop.

Sample Input:

Enter a number: 10

Expected Output:

2 4 6 8 10

Java Code:

```
import java.util.Scanner;

public class EvenNumbers {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        for (int i = 2; i <= n; i = i + 2) {
            System.out.print(i + " ");
        }

        sc.close();
    }
}
```

Output:

PS F:\Java Code\Logic Building\Assignment 4> javac EvenNumbers.java

PS F:\Java Code\Logic Building\Assignment 4> java EvenNumbers

Enter a number: 10

2 4 6 8 10

PS F:\Java Code\Logic Building\Assignment 4>

Question 5: Sum of Odd Numbers between 1 and N

Problem Statement:

Write a Java program that asks the user for a number N and calculates the sum of all odd numbers between 1 and N using a for loop.

Sample Input: Enter a number: 10

Expected Outcome: The sum of odd numbers from 1 to 10 is: 25

Java Code:

```
import java.util.Scanner;

public class SumOfOdd {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int n = sc.nextInt();
        int sum = 0;

        for (int i = 1; i <= n; i = i + 2) {
            sum = sum + i;
        }
        System.out.println("The sum of odd numbers from 1 to " + n + " is: " + sum);
        sc.close();
    }
}
```

Output:

PS F:\Java Code\Logic Building\Assignment 4> javac SumOfOdd.java

PS F:\Java Code\Logic Building\Assignment 4> java SumOfOdd

Enter a number: 10

The sum of odd numbers from 1 to 10 is: 25

PS F:\Java Code\Logic Building\Assignment 4>

Question 6: Print All Elements of an Array

Problem Statement:

Write a Java program that uses a for-each loop to print all elements of an integer array. The program should ask the user to input 5 integers, store them in an array, and then print all the elements using a for-each loop.

Sample Input: Enter 5 integers: 3 7 12 5 8

Expected Output: 3 7 12 5 8

Java Code:

```
import java.util.Scanner;

public class Array {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int[] numbers = new int[5];

        System.out.println("Enter 5 integers:");

        for (int i = 0; i < numbers.length; i++) {
            numbers[i] = sc.nextInt();
        }
        for (int number : numbers) {
            System.out.print(number + " ");
        }
        sc.close();
    }
}
```

Output:

PS F:\Java Code\Logic Building\Assignment 4> javac Array.java

PS F:\Java Code\Logic Building\Assignment 4> java Array

Enter 5 integers:

1 65 5 6 7

1 65 5 6 7

PS F:\Java Code\Logic Building\Assignment 4>

Question 7: Find the Sum of All Elements in an Array

Problem Statement:

Write a Java program that uses a for-each loop to calculate the sum of all elements in a given integer array. The program should ask the user to input 5 integers, store them in an array, and then compute the sum of these numbers using the for-each loop.

Sample Input: Enter 5 integers: 4 6 8 2 10

Expected Output: The sum of all numbers is: 30

Java Code:

```
import java.util.Scanner;

public class ArraySum {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int[] numbers = new int[5];

        System.out.println("Enter 5 integers:");

        for (int i = 0; i < numbers.length; i++) {
            numbers[i] = sc.nextInt();
        }

        int sum = 0;
        for (int number : numbers) {
            sum = sum + number;
        }
        System.out.println("The sum of all numbers is: " + sum);

        sc.close();
    }
}
```

Output:

PS F:\Java Code\Logic Building\Assignment 4> javac ArraySum.java

PS F:\Java Code\Logic Building\Assignment 4> java ArraySum

Enter 5 integers:

4 6 5 9 13 6

The sum of all numbers is: 97

PS F:\Java Code\Logic Building\Assignment 4>

Question 8: Print All Names in a String Array

Problem Statement:

Write a Java program that uses a for-each loop to print all the names stored in a String array. The program should ask the user to input 4 names, store them in an array, and then print each name using the for-each loop.

Sample Input: Enter 4 names: Manoj Aditya Shilbhushan Shweta

Expected Output: Manoj Aditya Shilbhushan Shweta

Java Code:

```
import java.util.Scanner;

public class ArrayName {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        String[] names = new String[4];

        System.out.println("Enter 4 names:");

        for (int i = 0; i < names.length; i++) {
            names[i] = sc.next();
        }

        for (String name : names) {
            System.out.println(name);
        }

        sc.close();
    }
}
```

Output:

PS F:\Java Code\Logic Building\Assignment 4> javac ArrayName.java

PS F:\Java Code\Logic Building\Assignment 4> java ArrayName

Enter 4 names:

Manoj Aditya Shilbhushan Shweta

Manoj

Aditya

Shilbhushan

Shweta

PS F:\Java Code\Logic Building\Assignment 4>

Question 9: Find the Largest Element in an Array

Problem Statement: Write a Java program that asks the user to input 5 integers, stores them in an array, and then finds and prints the largest element in the array. (Explore in-built method to solve this)

Sample Input: Enter 5 integers: 12 45 67 23 89

Expected Output: The largest element is: 89

Java Code:

```
import java.util.Arrays;
import java.util.Scanner;

public class LargestElement {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int[] numbers = new int[5];

        System.out.println("Enter 5 integers:");

        for (int i = 0; i < numbers.length; i++) {
            numbers[i] = sc.nextInt();
        }

        Arrays.sort(numbers);

        int largest = numbers[numbers.length - 1];

        System.out.println("The largest element is: " + largest);

        sc.close();
    }
}
```

Output:

PS F:\Java Code\Logic Building\Assignment 4> javac LargestElement.java

PS F:\Java Code\Logic Building\Assignment 4> java LargestElement

Enter 5 integers:

41 46 87 4 13

The largest element is: 87

PS F:\Java Code\Logic Building\Assignment 4>

Question 10: Find the Average of Elements in an Array

Problem Statement: Write a Java program that asks the user to input 5 integers, stores them in an array, and then calculates and prints the average of the elements in the array.

Sample Input: Enter 5 integers: 10 20 30 40 50

Expected Output: The average of the numbers is: 30.0

Java Code:

```
import java.util.Scanner;

public class ArrayAverage {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int[] arr = new int[5];
        int sum = 0;

        System.out.println("Enter 5 integers:");

        for (int i = 0; i < 5; i++) {
            arr[i] = sc.nextInt();
            sum = sum + arr[i];
        }

        double average = (double) sum / arr.length;

        System.out.println("The average of the numbers is: " + average);

        sc.close();
    }
}
```

Output:

PS F:\Java Code\Logic Building\Assignment 4> javac ArrayAverage.java

PS F:\Java Code\Logic Building\Assignment 4> java ArrayAverage

Enter 5 integers:

14 23 5 4 68

The average of the numbers is: 22.8

PS F:\Java Code\Logic Building\Assignment 4>

Question 11: Count Positive and Negative Numbers in an Array

Problem Statement: Write a Java program that asks the user to input 6 integers, stores them in an array, and then counts how many positive and negative numbers are present in the array.

Sample Input: Enter 6 integers: -5 3 7 -2 0 8

Expected Output: Positive numbers: 3 Negative numbers: 2

Java Code:

```
import java.util.Scanner;

public class PositiveNegative {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int[] arr = new int[6];
        int positive = 0;
        int negative = 0;
        System.out.println("Enter 6 integers:");

        for (int i = 0; i < 6; i++) {
            arr[i] = sc.nextInt();
        }
        for (int number : arr) {
            if (number > 0) {
                positive++;
            } else if (number < 0) {
                negative++;
            }
        }

        System.out.println("Positive numbers: " + positive);
        System.out.println("Negative numbers: " + negative);
        sc.close();
    }
}
```

Output:

PS F:\Java Code\Logic Building\Assignment 4> javac PositiveNegative.java

PS F:\Java Code\Logic Building\Assignment 4> java PositiveNegative

Enter 6 integers:

5 -14 23 10 15 -18

Positive numbers: 4

Negative numbers: 2

Question 12: Sort an Array in Ascending Order

Problem Statement: Write a Java program that asks the user to input 5 integers, stores them in an array, and then sorts the array in ascending order using the Arrays.sort() method. After sorting, print the sorted array.

Sample Input: Enter 5 integers: 12 45 23 8 90

Expected Output: Sorted array: 8 12 23 45 90

Java Code:

```
import java.util.Arrays;
import java.util.Scanner;

public class SortArray {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int[] arr = new int[5];

        System.out.println("Enter 5 integers:");

        for (int i = 0; i < 5; i++) {
            arr[i] = sc.nextInt();
        }

        Arrays.sort(arr);

        System.out.print("Sorted array: ");

        for (int number : arr) {
            System.out.print(number + " ");
        }

        sc.close();
    }
}
```

Output:

PS F:\Java Code\Logic Building\Assignment 4> javac SortArray.java

PS F:\Java Code\Logic Building\Assignment 4> java SortArray

Enter 5 integers:

16 6 5 14 7

Sorted array: 5 6 7 14 16

PS F:\Java Code\Logic Building\Assignment 4>

Question 13: Check if an Array Contains a Specific Element

Problem Statement:

Write a Java program that asks the user to input 5 integers, stores them in an array, and then checks whether a specific number (input by the user) is present in the array using the `Arrays.asList()` method. If the number is found, print "Found", otherwise print "Not Found".

Sample Input: Enter 5 integers: 10 20 30 40 50 Enter the number to search: 30

Expected Output: Found

Java Code:

```
import java.util.Arrays;
import java.util.Scanner;

public class SearchElement {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        Integer[] arr = new Integer[5];
        System.out.println("Enter 5 integers:");

        for (int i = 0; i < 5; i++) {
            arr[i] = sc.nextInt();
        }
        System.out.print("Enter the number to search: ");
        int number = sc.nextInt();

        if (Arrays.asList(arr).contains(number)) {
            System.out.println("Found");
        } else {
            System.out.println("Not Found");
        }
        sc.close();
    }
}
```

Output:

PS F:\Java Code\Logic Building\Assignment 4> javac SearchElement.java

PS F:\Java Code\Logic Building\Assignment 4> java SearchElement

Enter 5 integers:

4 6 5 7 9

Enter the number to search: 5

Found

PS F:\Java Code\Logic Building\Assignment 4> java SearchElement

Enter 5 integers:

5 6 7 3 1

Enter the number to search: 2

Not Found

Question 14: Find the Index of an Element in an Array

Problem Statement: Write a Java program that asks the user to input 5 integers, stores them in an array, and then finds the index of a specific number (input by the user) using the `Arrays.binarySearch()` method. If the number is found, print the index, otherwise print "Not Found".

Sample Input: Enter 5 integers: 5 10 15 20 25 Enter the number to search: 15

Expected Output: The number 15 is found at index 2

Java Code:

```
import java.util.Arrays;
import java.util.Scanner;

public class Index {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int[] arr = new int[5];

        System.out.println("Enter 5 integers:");

        for (int i = 0; i < 5; i++) {
            arr[i] = sc.nextInt();
        }

        System.out.print("Enter the number to search: ");
        int number = sc.nextInt();

        int index = Arrays.binarySearch(arr, number);

        if (index >= 0) {
            System.out.println("The number " + number +
                               " is found at index " + index);
        } else {
            System.out.println("Not Found");
        }
        sc.close();
    }
}
```

Output:

PS F:\Java Code\Logic Building\Assignment 4> javac Index.java

PS F:\Java Code\Logic Building\Assignment 4> java Index

Enter 5 integers:

1 3 5 6 4

Enter the number to search: 6

The number 6 is found at index 3

Question 15:Write a program to print the following pattern:

```
1
2*2
3*3*3
4*4*4*4
5*5*5*5*5
5*5*5*5*5
4*4*4*4
3*3*3
2*2
```

Java Code:

```
public class Pattern {
    public static void main(String[] args) {

        for (int i = 1; i <= 5; i++) {
            for (int j = 1; j <= i; j++) {
                System.out.print(i);

                if (j < i) {
                    System.out.print("*");
                }
            }
            System.out.println();
        }

        for (int i = 5; i >= 2; i--) {
            for (int j = 1; j <= i; j++) {
                System.out.print(i);

                if (j < i) {
                    System.out.print("*");
                }
            }
            System.out.println();
        }
    }
}
```

Output:

PS F:\Java Code\Logic Building\Assignment 4> javac Pattern.java

PS F:\Java Code\Logic Building\Assignment 4> java Pattern

1

2*2

3*3*3

4*4*4*4

5*5*5*5*5

5*5*5*5*5

4*4*4*4

3*3*3

2*2

Question 16: Write a program to print the following pattern:

```
1
1*2
1*2*3
1*2*3*4
1*2*3*4*5
```

Java Code:

```
public class Pattern16 {
    public static void main(String[] args) {

        for (int i = 1; i <= 5; i++) {

            for (int j = 1; j <= i; j++) {
                System.out.print(j);

                if (j < i) {
                    System.out.print("*");
                }
            }

            System.out.println();
        }
    }
}
```

Output:

PS F:\Java Code\Logic Building\Assignment 4> javac Pattern16.java

PS F:\Java Code\Logic Building\Assignment 4> java Pattern16

```
1
1*2
1*2*3
1*2*3*4
1*2*3*4*5
```

PS F:\Java Code\Logic Building\Assignment 4>

Question 17: Write a program to print the following pattern:

```
1
1*3
1*3*5
1*3*5*7
1*3*5*7*9
```

Java Code:

```
public class Pattern17 {
    public static void main(String[] args) {

        for (int i = 1; i <= 5; i++) {

            for (int j = 1; j <= i; j++) {
                int number = (2 * j) - 1;

                System.out.print(number);

                if (j < i) {
                    System.out.print("*");
                }
            }

            System.out.println();
        }
    }
}
```

Output:

PS F:\Java Code\Logic Building\Assignment 4> javac Pattern17.java

PS F:\Java Code\Logic Building\Assignment 4> java Pattern17

```
1
1*3
1*3*5
1*3*5*7
```

Question 18: Write a program to print the following pattern:

```
11111
22222
33333
44444
55555
```

Java Code:

```
public class Pattern18 {
    public static void main(String[] args) {

        for (int i = 1; i <= 5; i++) {
            for (int j = 1; j <= 5; j++) {
                System.out.print(i);
            }

            System.out.println();
        }
    }
}
```

Output:

PS F:\Java Code\Logic Building\Assignment 4> javac Pattern18.java

PS F:\Java Code\Logic Building\Assignment 4> java Pattern18

```
11111
22222
33333
44444
55555
```

Question 19: Write a program to print the following pattern:

```
1
22
333
4444
55555
```

Java Code:

```
public class Pattern19 {
    public static void main(String[] args) {

        for (int i = 1; i <= 5; i++) {
            for (int j = 1; j <= i; j++) {
                System.out.print(i);
            }

            System.out.println();
        }
    }
}
```

Output:

PS F:\Java Code\Logic Building\Assignment 4> javac Pattern19.java

PS F:\Java Code\Logic Building\Assignment 4> java Pattern19

```
1
22
333
4444
55555
```

Question 20: Write a program to print the following pattern:

```
1
12
123
1234
12345
```

Java Code:

```
public class Pattern20 {
    public static void main(String[] args) {

        for (int i = 1; i <= 5; i++) {
            for (int j = 1; j <= i; j++) {
                System.out.print(j);
            }

            System.out.println();
        }
    }
}
```

Output:

```
PS F:\Java Code\Logic Building\Assignment 4> javac Pattern20.java
PS F:\Java Code\Logic Building\Assignment 4> java Pattern20
```

```
1
12
123
1234
12345
```

Question 21: Write a program to print the following pattern:

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

Java Code:

```
public class Pattern21 {
    public static void main(String[] args) {

        int number = 1;

        for (int i = 1; i <= 5; i++) {

            for (int j = 1; j <= i; j++) {
                System.out.print(number + " ");
                number++;
            }

            System.out.println();
        }
    }
}
```

Output:

PS F:\Java Code\Logic Building\Assignment 4> javac Pattern21.java

PS F:\Java Code\Logic Building\Assignment 4> java Pattern21

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

Question 22: Write a program to print the following pattern:

```
*****
*      *
*      *
*      *
*      *
*      *
*****
```

Java Code:

```
public class Pattern22 {
    public static void main(String[] args) {

        int rows = 6;
        int columns = 6;

        for (int i = 1; i <= rows; i++) {

            for (int j = 1; j <= columns; j++) {

                if (i == 1 || i == rows || j == 1 || j == columns) {
                    System.out.print("*");
                } else {
                    System.out.print(" ");
                }
            }

            System.out.println();
        }
    }
}
```

Output:

PS F:\Java Code\Logic Building\Assignment 4> javac Pattern22.java

PS F:\Java Code\Logic Building\Assignment 4> java Pattern22

```
*****
*      *
*      *
*      *
*      *
*****
```

Question 23: Write a program to print the following pattern:



```
*
***
*****
*****
*****
*****
*****
*****
*****
*****
*****
*
```

Java Code:

```
public class Pattern23 {
    public static void main(String[] args) {

        // Upper Side
        for (int i = 1; i <= 6; i++) {

            for (int space = 1; space <= 6 - i; space++) {
                System.out.print(" ");
            }
            for (int star = 1; star <= (2 * i - 1); star++) {
                System.out.print("*");
            }
            System.out.println();
        }

        // Lower side
        for (int i = 5; i >= 1; i--) {

            for (int space = 1; space <= 6 - i; space++) {
                System.out.print(" ");
            }
            for (int star = 1; star <= (2 * i - 1); star++) {
                System.out.print("*");
            }
            System.out.println();
        }
    }
}
```


Output:

```
PS F:\Java Code\Logic Building\Assignment 4> javac Pattern23.java
```

PS F:\Java Code\Logic Building\Assignment 4> java Pattern23

```

      *
    ***
  *****
*****
*****
*****
*****
*****
*****
*****
*****

```

Question 24: Reverse a String

Problem Statement:

Write a Java program that asks the user for a string and then prints the reverse of that string.

Sample Input: Enter a string: hello

Expected Output: Reversed string: olleh

Java Code:

```
import java.util.Scanner;

public class ReverseString {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a string: ");
        String text = sc.nextLine();

        String reverse = "";

        for (int i = text.length() - 1; i >= 0; i--) {
            reverse = reverse + text.charAt(i);
        }

        System.out.println("Reversed string: " + reverse);

        sc.close();
    }
}
```

Output:

PS F:\Java Code\Logic Building\Assignment 4> javac ReverseString.java

PS F:\Java Code\Logic Building\Assignment 4> java ReverseString

Enter a string: Hello

Reversed string: olleH

Question 25: Count Vowels in a String

Problem Statement:

Write a Java program that asks the user for a string and counts the number of vowels (a, e, i, o, u) in the string. The program should then print the total number of vowels.

Sample Input: Enter a string: programming

Expected Output: The number of vowels in 'programming' is: 3

Java Code:

```
import java.util.Scanner;

public class Vowels {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a string: ");
        String text = sc.nextLine();

        int count = 0;
        for (int i = 0; i < text.length(); i++) {
            char ch = Character.toLowerCase(text.charAt(i));
            if (ch == 'a' || ch == 'e' || ch == 'i' ||
                ch == 'o' || ch == 'u') {
                count++;
            }
        }
        System.out.println("The number of vowels in '" + text + "' is: " + count);
        sc.close();
    }
}
```

Output:

PS F:\Java Code\Logic Building\Assignment 4> javac Vowels.java

PS F:\Java Code\Logic Building\Assignment 4> java Vowels

Enter a string: programming

The number of vowels in 'programming' is: 3

Question 26: Check if a String is a Palindrome

Problem Statement:

Write a Java program that asks the user for a string and checks whether the string is a palindrome. A palindrome is a string that reads the same backward as forward (ignoring spaces and punctuation).

Sample Input: Enter a string: madam

Expected Output: The string 'madam' is a palindrome.

Java Code:

```
import java.util.Scanner;

public class Palindrome {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a string: ");
        String text = sc.nextLine();

        String cleanText = text.replaceAll("[^a-zA-Z0-9]", "");
        String reverse = "";
        for (int i = cleanText.length() - 1; i >= 0; i--) {
            reverse = reverse + cleanText.charAt(i);
        }
        if (cleanText.equalsIgnoreCase(reverse)) {
            System.out.println("The string '" + text + "' is a palindrome.");
        } else {
            System.out.println("The string '" + text + "' is not a palindrome.");
        }
        sc.close();
    }
}
```

Output:

PS F:\Java Code\Logic Building\Assignment 4> javac Palindrome.java

PS F:\Java Code\Logic Building\Assignment 4> java Palindrome

Enter a string: madam

The string 'madam' is a palindrome.

Question 27: String Literal and Object Creation

Problem Statement:

Write a Java program that creates two string variables using string literals with the same content. Then, print whether both variables point to the same object.

Code Example: String str1 = "hello"; String str2 = "hello";

Expected Output: Both variables point to the same object: true

Java Code:

```
public class StringLiteral {  
    public static void main(String[] args) {  
  
        String str1 = "hello";  
        String str2 = "hello";  
  
        System.out.println(  
            "Both variables point to the same object: " + (str1 == str2)  
        );  
    }  
}
```

Output:

PS F:\Java Code\Logic Building\Assignment 4> javac StringLiteral.java

PS F:\Java Code\Logic Building\Assignment 4> java StringLiteral

Both variables point to the same object: true

Question 28: String Creation with new Keyword

Problem Statement:

Write a Java program that creates two string objects using the new keyword with the same content. Then, print whether both objects are the same using the == operator and the .equals() method.

Code Example: String str1 = new String("hello"); String str2 = new String("hello");

Expected Output: Using == : false Using .equals(): true

Java Code:

```
public class StringObject {  
    public static void main(String[] args) {  
  
        String str1 = new String("hello");  
        String str2 = new String("hello");  
  
        System.out.println("Using == : " + (str1 == str2));  
        System.out.println("Using .equals(): " + str1.equals(str2));  
    }  
}
```

Output:

PS F:\Java Code\Logic Building\Assignment 4> javac StringObject.java

PS F:\Java Code\Logic Building\Assignment 4> java StringObject

Using == : false

Using .equals(): true

PS F:\Java Code\Logic Building\Assignment 4>

Question 29: String Concatenation and Object Creation

Problem Statement:

Write a Java program that concatenates two strings using the + operator. Print whether the concatenated string is a new object or a reference to an existing string object using the == operator.

Code Example: String str1 = "hello"; String str2 = "world"; String str3 = str1 + str2;

Expected Output: Is str3 pointing to the same object as str1? false

Java Code:

```
public class StringConcatenation {  
    public static void main(String[] args) {  
  
        String str1 = "hello";  
        String str2 = "world";  
  
        String str3 = str1 + str2;  
  
        System.out.println(  
            "Is str3 pointing to the same object as str1? " + (str3 == str1)  
        );  
    }  
}
```

Output:

PS F:\Java Code\Logic Building\Assignment 4> javac StringConcatenation.java

PS F:\Java Code\Logic Building\Assignment 4> java StringConcatenation

Is str3 pointing to the same object as str1? false

PS F:\Java Code\Logic Building\Assignment 4>

Question 30: String Pool with intern() Method

Problem Statement:

Write a Java program that creates a string using the new keyword and then calls the intern() method. Print whether the interned string is pointing to the same object as the original string literal.

Code Example: String str1 = new String("hello"); String str2 = str1.intern(); String str3 = "hello";

Expected Output: Is str2 and str3 pointing to the same object? true

Java Code:

```
public class StringIntern {  
    public static void main(String[] args) {  
  
        String str1 = new String("hello");  
  
        String str2 = str1.intern();  
  
        String str3 = "hello";  
  
        System.out.println(  
            "Is str2 and str3 pointing to the same object? " + (str2 == str3)  
        );  
    }  
}
```

Output:

PS F:\Java Code\Logic Building\Assignment 4> javac StringIntern.java

PS F:\Java Code\Logic Building\Assignment 4> java StringIntern

Is str2 and str3 pointing to the same object? true

PS F:\Java Code\Logic Building\Assignment 4>

Question 31: Multiple String Literals with Same Content

Problem Statement:

Write a Java program that declares three string literals with the same content and prints whether all three strings refer to the same object using the == operator.

Code Example: String str1 = "java"; String str2 = "java"; String str3 = "java";

Expected Output: All strings point to the same object: true

Java Code:

```
public class MultipleStrings {  
    public static void main(String[] args) {  
  
        String str1 = "java";  
        String str2 = "java";  
        String str3 = "java";  
  
        boolean result = (str1 == str2) && (str2 == str3);  
  
        System.out.println("All strings point to the same object: " + result);  
    }  
}
```

Output:

PS F:\Java Code\Logic Building\Assignment 4> javac MultipleStrings.java

PS F:\Java Code\Logic Building\Assignment 4> java MultipleStrings

All strings point to the same object: true

PS F:\Java Code\Logic Building\Assignment 4>